ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

TASK 8 – PLAN THE ALGORITHM BEFORE CODING

1. Algorithm

The algorithm is a step-by-step procedure used to solve a problem. In the milk payment calculator, the algorithm calculates the milk rate and the total payment based on the quantity of milk, fat percentage, and SNF percentage.

Step 1: Start the program.

Step 2: Define the base rate as ₹30 per litre, base fat as 3.5%, and base SNF as 8.5%.

Step 3: Get the farmer ID, milk quantity, fat percentage, and SNF percentage.

Step 4: Calculate the fat difference and SNF difference from their base values.

Step 5: Calculate the number of 0.1% steps for fat and SNF.

Step 6: Calculate the milk rate using the base rate and the fat and SNF adjustments.

Step 7: Calculate the total payment by multiplying the milk quantity by the rate per litre.

Step 8: Round the payment amount to two decimal places.

Step 9: Display the farmer details, milk rate, and total payment.

Step 10: Stop the program.

2. Importance of Planning Before Coding

Planning the algorithm before coding helps us understand the problem clearly and arrange the calculations in the correct order. It reduces logical errors, makes the program easier to test, and improves readability and maintenance.

3. Alternative Approaches

The algorithm can be represented using pseudocode or a flowchart. Pseudocode explains the steps using simple statements, while a flowchart represents the steps using symbols and arrows. Both methods help us understand the program logic before implementation.

4. Time and Space Complexity

The algorithm performs a fixed number of calculations for one farmer. Therefore, the time complexity is O(1), and the auxiliary space complexity is also O(1), because the program uses a fixed number of variables.

